



East West University

Department of Computer Science and Engineering

Semester : Spring 2021

Course Title : Electrical Circuits

Course Code : CSE209

Section : 03

Name of project :

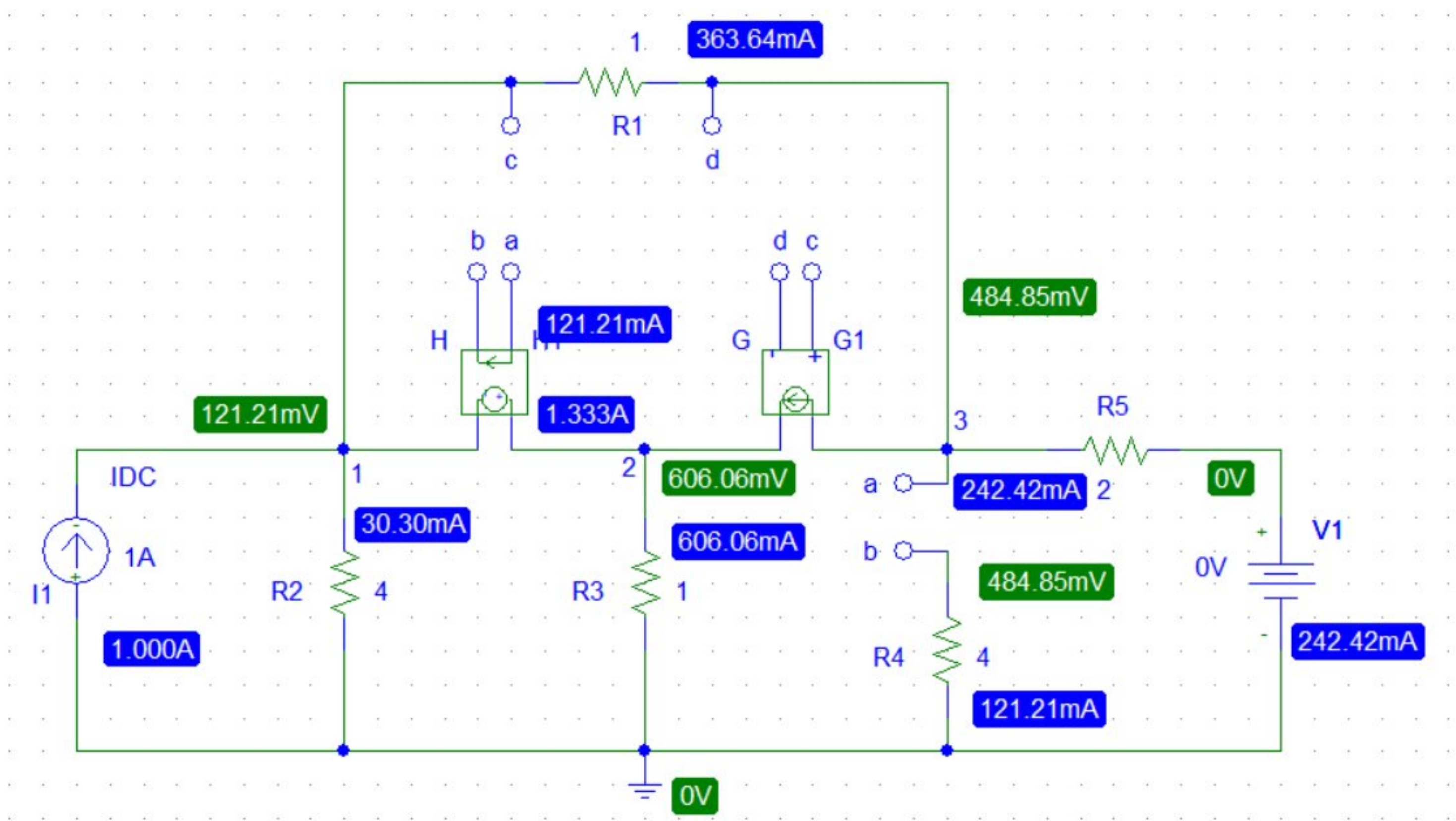
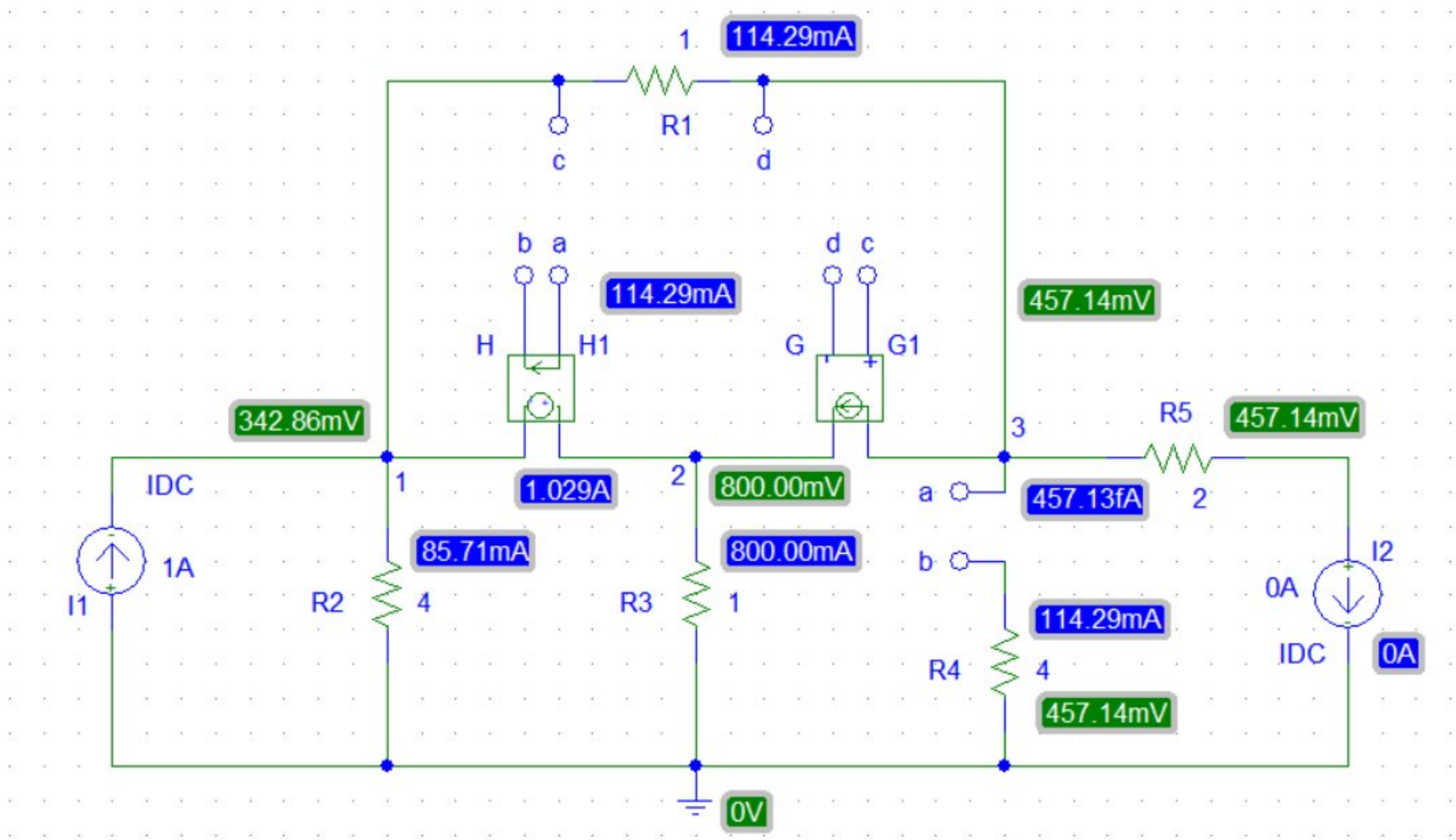
**PSpice Analysis for
Maximum Power Transfer**

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Circuit Diagram :



From PSpice circuit diagrams,

$$V_{th} = V_{oc} = 0.4571V$$

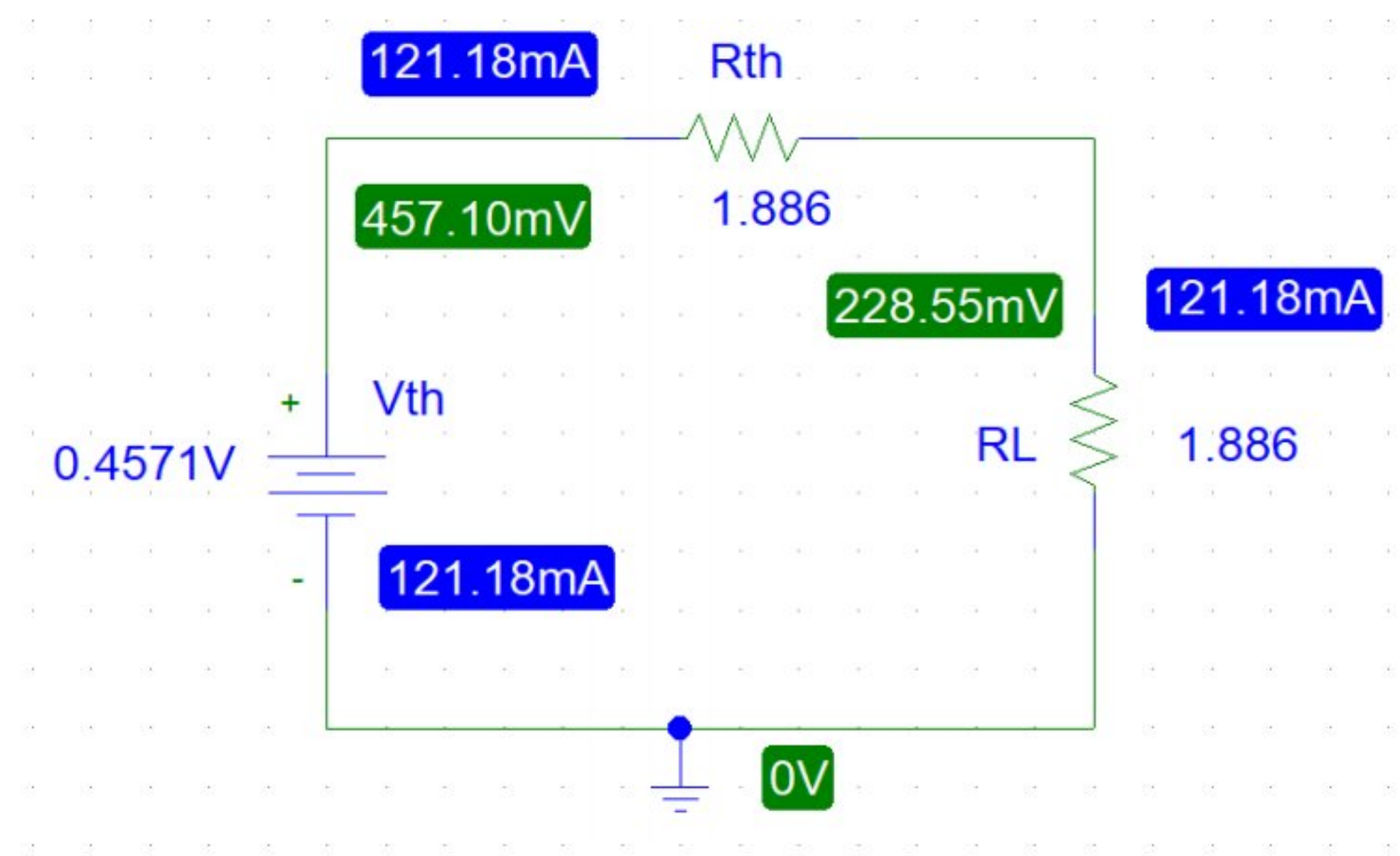
$$I_{sc} = 0.2424A$$

$$\begin{aligned} R_{th} &= \frac{V_{th}}{I_{sc}} = \frac{0.4571}{0.2424} = 1.886\Omega \end{aligned}$$

For maximum power transfer, $R_L = R_{th}$

$$R_L = 1.886\Omega$$

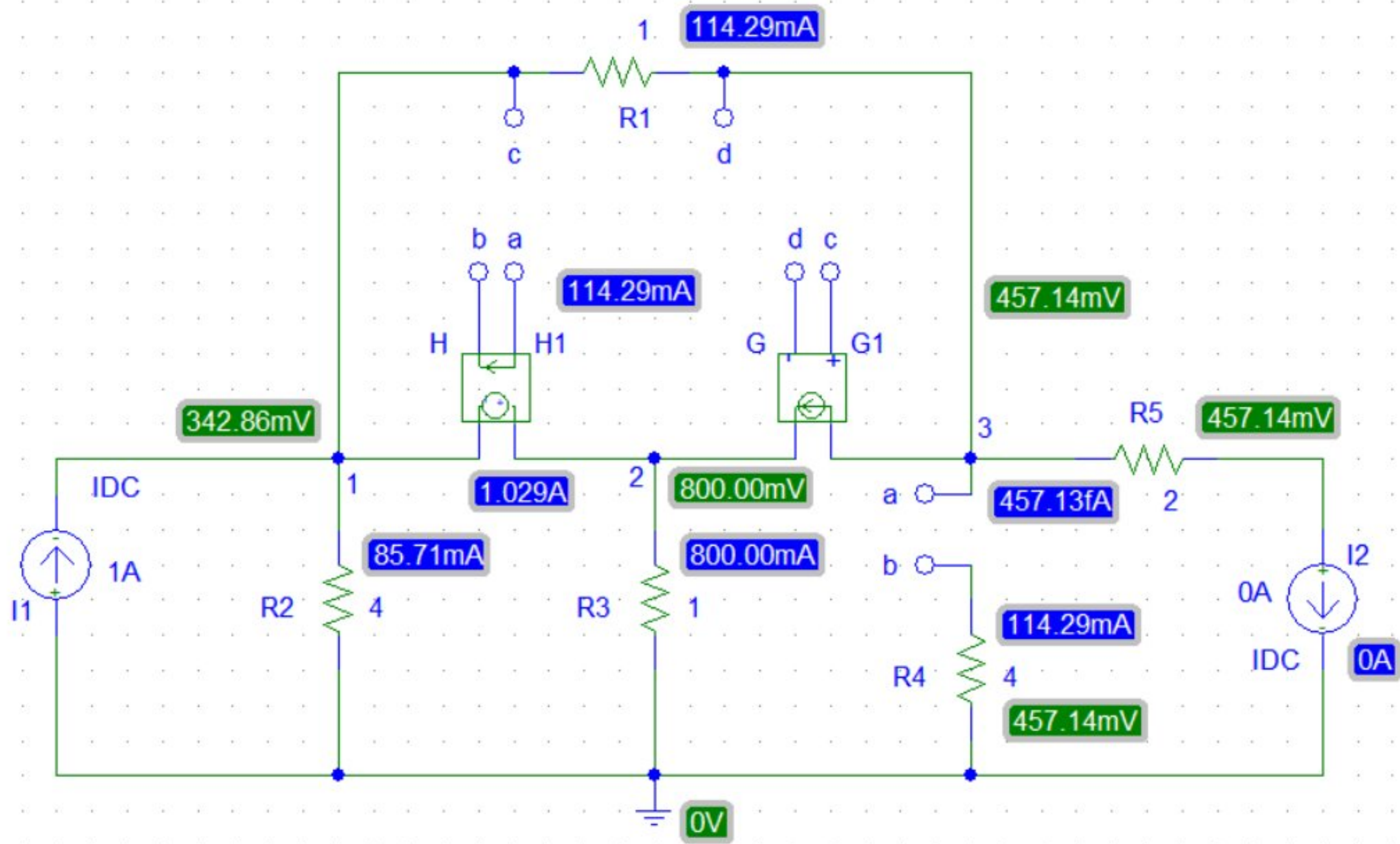
$$\begin{aligned} P_{max} &= \frac{(V_{th})^2}{4 \cdot R_{th}} = \frac{(0.4571)^2}{4 \cdot 1.886} \\ &= 0.02769W = 27.696mw \end{aligned}$$



Thevenin's equivalent of the circuit

Theoretical solution:

Vth :



Applying KCL at supernode :

$$-3V_1 + 4V_2 + 4V_3 = 4 \dots (1)$$

From supernode :

$$-V_1 + V_2 - V_3 = 0 \dots (2)$$

Applying KCL at node V3 :

$$4V_1 - 3V_3 = 0 \dots (3)$$

Solving (1) , (2) , (3) we get ,

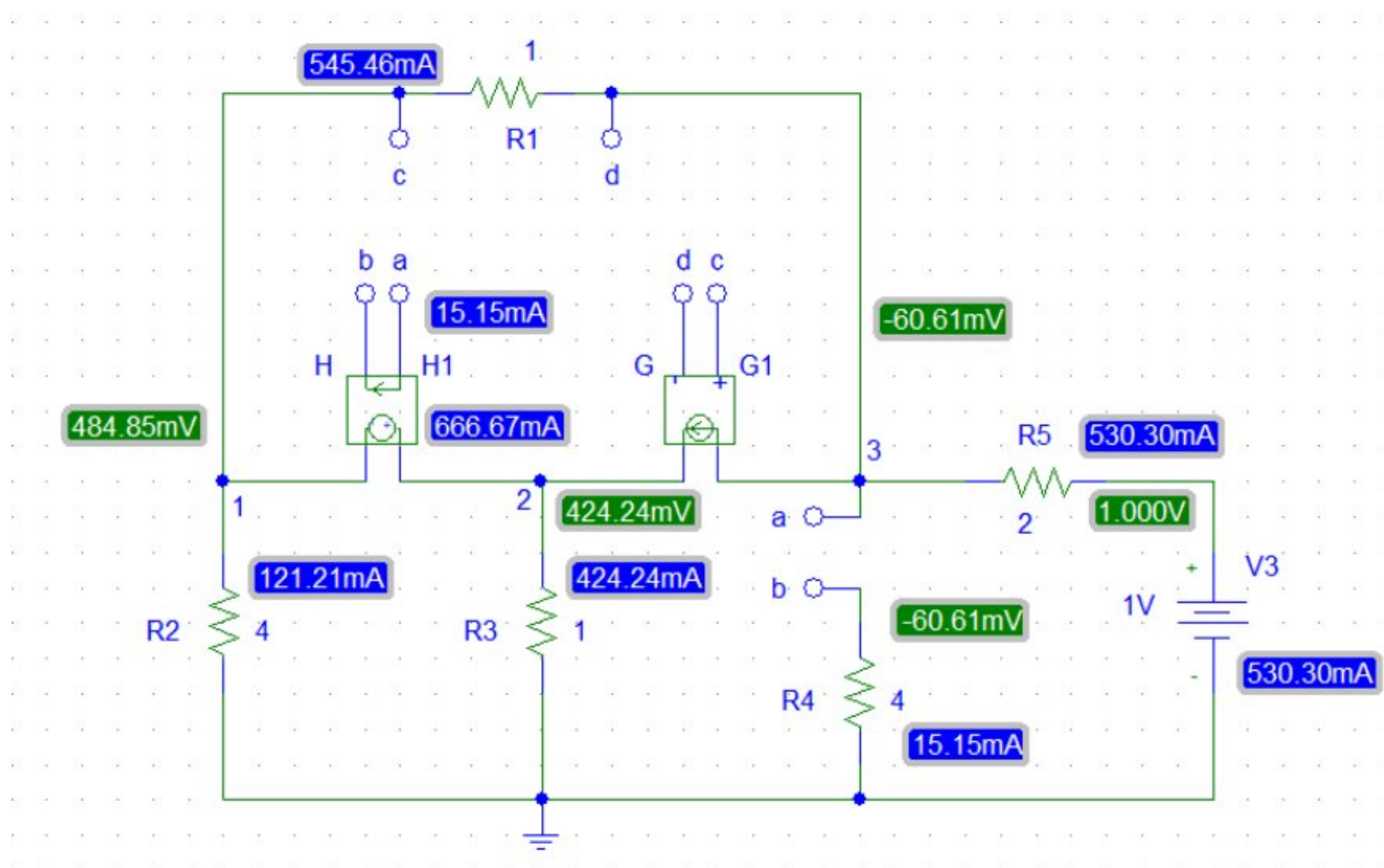
$$V_1 = 0.3428V$$

$$V_2 = 0.8V$$

$$V_3 = V_{oc} = 0.4571V$$

$$V_{th} = 0.4571V$$

R_{th} :



Applying KCL at Supernode:

$$-3V_1 + 4V_2 + 4V_3 = 0 \quad (1)$$

From Supernode:

$$-V_1 + V_2 - V_3 = 0 \quad (2)$$

Applying KCL at node V3 :

$$4V_1 - V_3 = 2 \dots (3)$$

Solving (1) , (2) ,(3) we get ,

$$V_1 = 0.485V$$

$$V_2 = 0.424V$$

$$V_3 = -0.060V$$

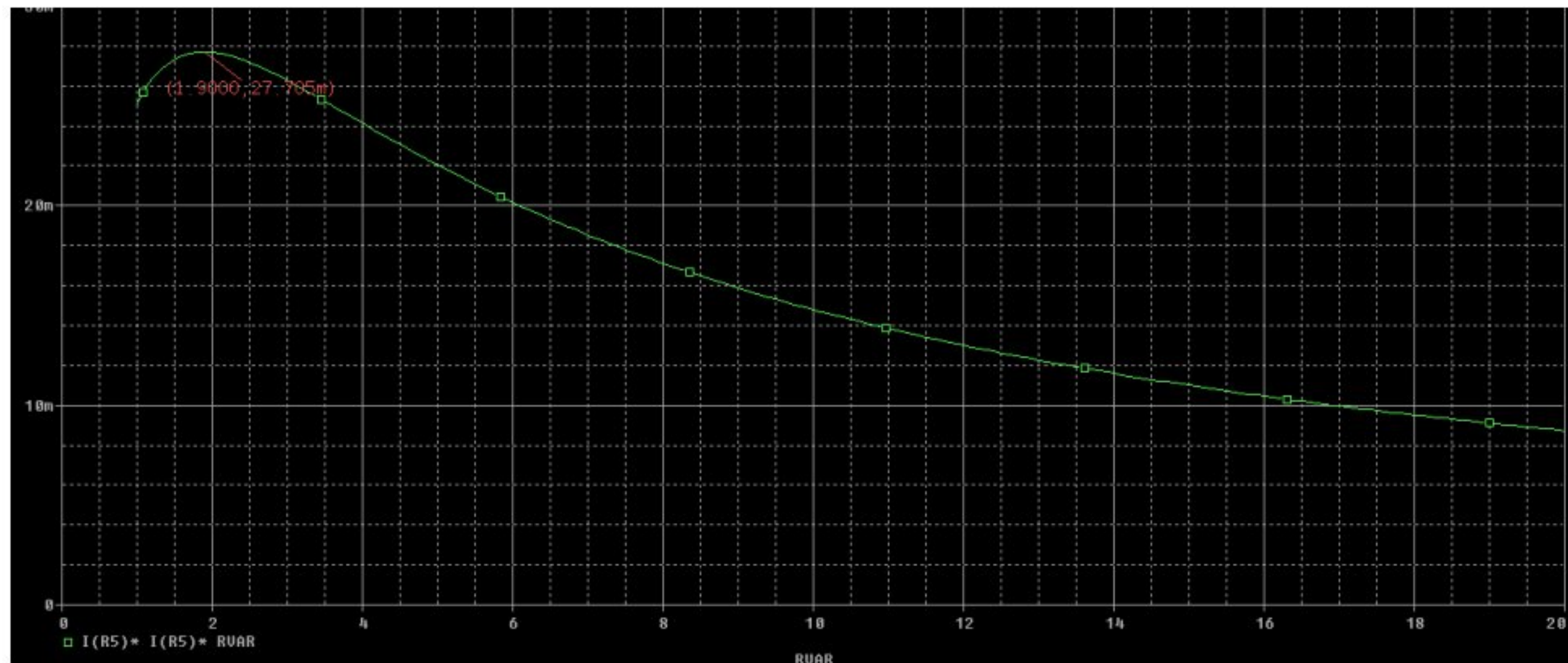
$$I_o = 1 - V_3 / 2 = 1 - (-0.060) / 2 = 0.53 A$$

$$R_{th} = v_0 / i_0 = 1 / 0.53 = 1.886\Omega$$

For maximum power transfer , $R_L = R_{th}$

$$R_L = 1.886\Omega$$

$$P_{max} = \frac{(V_{th})^2}{4 \cdot R_{th}} = \frac{(0.4571)^2}{4 \cdot 1.886} = 0.02769W = 27.696mW$$



From PSpice Simulation with resistance sweep,

$$R_L = 1.9\Omega$$

$$P_{max} = 27.705mW$$

Comparison between the values of R_L and P_{max} obtained in steps 2 and 3.

	Step 2	Step 3
R_L	1.886Ω	1.9Ω
P_{max}	$27.696mW$	$27.705mW$

Comment: Values of R_L and P_{max} obtained in steps 2 and 3 are pretty much the same.

Conclusion :

The main objective of the project was to analyze the maximum power transfer using PSpice. After comparing the theoretically obtained values and PSpice measured values of R_L and P_{max} , it is quite visible that both the values of R_L and P_{max} are almost same.

